

EPISODE - 03

Does ChatGPT Know or Does It Guess?

Complete Episode 3 Notes

THE BIG QUESTION

Does ChatGPT actually know things, or is it predicting what comes next? If it is predicting, why can the answer sometimes be correct and useful, but sometimes confidently wrong?

Search engines retrieve. LLMs generate. Modern AI assistants combine generation with tools, retrieval, instructions, context and other capabilities.

1. Search Engines vs. LLMs

A search engine is primarily a retrieval system.

- Google creates an index of web pages. Think of it like the index at the back of a physics book.
- If you want to find "thermodynamics" in a physics book, you use the index to find the relevant chapter/page, then read it.
- A search engine similarly uses its index to find relevant web pages for your query.
- It ranks candidate documents/pages and returns results it considers relevant.

Simple search flow:



Search engines = RETRIEVE information.

ChatGPT / LLM

- You give a prompt.
- The model has learned patterns from large amounts of training data.
- It predicts the next token/text based on the prompt and context.
- It keeps generating tokens until it forms a response.

LLMs = GENERATE information.

MAIN DIFFERENCE

Search Engine -> Retrieve LLM -> Generate Search can point you toward existing documents. An LLM generates a new sequence of tokens using learned patterns.

2. Search Results Are Not Automatically Truth

- Search results can be outdated.
- Ranking can be imperfect.
- Results can be misleading.
- A high ranking does not automatically mean a page is authoritative or correct.

Good verification habits:

- Follow the trail back to the original source.
- Check who published it.
- Check the publication/update date.
- Compare multiple results/sources.

KEY IDEA

Search gives you documents to inspect. You still need to evaluate the evidence.

3. How LLMs Generate Responses

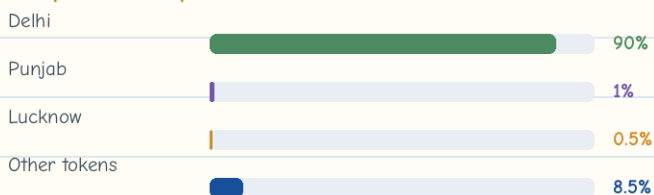
The basic idea is next-token prediction.

Example: The sun rises in the ____.

- The model predicts likely next tokens from the context.
- For a simple continuation, "east" is highly likely because the model has learned that pattern from training data.

An LLM does not simply pick one word blindly. It can represent many possible next tokens with different probabilities.

Example: The capital of India is ____



IMPORTANT

The percentages above are illustrative. The point is that generation comes from a probability distribution over possible next tokens.

4. Is an LLM Just Random Guessing?

Calling a modern LLM simply a random word guesser or just an autocomplete understates what the technology can do.

- Modern LLMs use much larger context.

- They have learned extremely complex patterns.
- Patterns can include grammar, programming, reasoning patterns, facts, languages, stories, mathematics, and associations between people, places, events and ideas.

BETTER MENTAL MODEL

LLM generation is probabilistic, but it is not arbitrary random guessing. The probabilities are shaped by patterns learned during training and by the current context.

5. Knowledge Cutoff

Every LLM has a knowledge cutoff associated with its training data.

- Models are not automatically learning new things every day.
- Neural networks are not normally retrained every day.
- The model is not automatically real-time just because you are chatting with it.
- Information after the training cutoff may not be present in the model's learned parameters.

REMEMBER

A training cutoff is one reason an LLM may not know a recent event. Current information can instead come from web search, APIs, databases, or other retrieval systems.

6. Base Model vs. AI Assistant

Base model: a model trained primarily to predict text/tokens.

Examples:

- "Once upon a time ..." -> predict what comes next.
- "The JS Event Loop is ..." -> predict a likely continuation.
- "User: What is Namaste AI? Assistant: ..." -> generate a continuation.

But ChatGPT is not simply the raw base model.

An AI assistant can place many capabilities around the underlying model:

- Web search
- Files
- Instruction tuning
- Memory
- Retrieval
- Tool access
- System instructions
- Safety training
- Conversation management
- Content filters
- Security / Auth

CAR + ENGINE ANALOGY

Car = ChatGPT / AI Assistant Engine = Base Model The engine is the core mechanism, but the complete car has many other systems that make it useful.

7. Training vs. Inference

Training happens before normal use.

- A neural network is created.
- Large amounts of content are used during training.
- The model has a huge collection of parameters/weights.
- Training repeatedly adjusts those parameters so the model learns useful patterns.

Inference happens after training.

- A user provides a prompt.
- The trained model processes the prompt/context.
- It generates a response from the trained model.
- The act of producing the response is called inference.



ONE-LINE DEFINITION

Training = learning/adjusting the model. Inference = using the trained model to generate a response.

8. Why Models Confidently Produce Wrong Answers

Fake fluency != truthfulness

An answer can be fluent, well structured, and confident while still being factually wrong.

HALLUCINATION

A hallucination occurs when an AI model generates information that appears plausible but is unsupported, incorrect, misleading, or fabricated.

Language quality and factual accuracy are separate dimensions.

Confidence in language is not the same as confidence in truth.

Why hallucinations happen:

- Insufficient information
- Ambiguous information
- Outdated knowledge
- False assumptions
- Unreliable learned patterns
- Models are optimized to answer rather than automatically detect truth
- Generation is probabilistic

9. Types of Hallucinations

- Invented facts -> The model creates a fact or event that does not exist.
- Invented citations -> The model produces a paper, source, link, quote, or reference that is fabricated.
- Incorrect combinations -> Correct pieces of information are combined in an incorrect way.
- Outdated facts -> The answer reflects information that is no longer current.
- False precision -> The model gives an exact number/date/detail without reliable support.
- Broken reasoning -> The final conclusion or reasoning chain is logically incorrect.

10. Why Do Models Sometimes Say "I Don't Know"?

A model can be trained or instructed to acknowledge uncertainty instead of fabricating an answer.

- Assistant training can encourage helpful uncertainty and refusal to invent facts.
- System instructions can tell the assistant to say "I don't know" when information is unavailable.
- Weak patterns may make the model less willing to confidently continue.
- Safety rules can discourage unsupported or risky answers.
- Tool requirements can make the system recognize that web search, a calculator, a database, or another tool is needed.
- Prompt wording can influence whether the assistant attempts an answer or expresses uncertainty.

IMPORTANT

Saying "I don't know" does not prove the model is conscious of its ignorance. It can be the result of training, instructions, context, tools, and learned response behaviour.

11. The Confidence Illusion

Humans often use tone as evidence. With LLMs, that can be dangerous.

"I think the answer may be XYZ."

"The answer is definitely XYZ."

Both statements can still be wrong.

LLMs can produce confident-sounding mistakes.

Good verification habits:

- Separate facts from assumptions.
- Ask for uncertainty when appropriate.
- Ask for sources.
- Use web search or other tools when current/verified information is needed.
- Verify important claims independently.

12. How Tools Extend the Model

Tools act like additional capabilities or "superpowers" around the model.

- Web search -> current external information
- Calculator -> accurate calculations
- Code execution -> run/test code
- Weather -> current/forecast weather
- Location -> location-based information
- Calendar -> schedules and events
- Email -> interact with email
- Database -> structured data
- Internal docs -> organization-specific knowledge
- Files -> read/analyze documents
- And much more...

CORE IDEA

Base Model -> generates language Base Model + Tools -> can retrieve information, perform calculations, run code, access data, and interact with external systems. Tools do not completely eliminate errors; they extend the model's capabilities.

13. Web Search + LLM = RAG

RAG = Retrieval-Augmented Generation

Retrieval gives external evidence. Generation converts it into a useful response.



- Retrieval -> find relevant external information.
- Generation -> use retrieved context to produce a natural-language response.

RAG IN ONE LINE

Retrieval + Generation = Retrieval-Augmented Generation (RAG)

NOTE: Tools and retrieval can reduce errors, but they do not completely eliminate hallucinations.

14. Does the Model Know Itself? - Self-Awareness

An LLM can talk about itself, but that is not the same thing as proving human-like consciousness or self-awareness.

- Training -> learned patterns about language and the world, including patterns about AI assistants.
- Context -> information supplied in the current conversation.
- System instructions -> rules about how the assistant should behave and respond.
- Tools -> external information or capabilities available during a task.

IMPORTANT DISTINCTION

If an AI says "I am ChatGPT," it can be generating that statement from its training, instructions, context, and system design. Self-description is not the same as human-like self-awareness.

15. Complete Mental Model

Put the whole episode together:



Around the base model, an AI assistant can add:

Instructions + Context + Tools + Retrieval + Safety + Memory/State + Other systems

FINAL TAKEAWAY

ChatGPT is not simply a search engine. A search engine primarily retrieves documents from an index, while an LLM generates text from learned patterns. LLMs are not just random guessing. Their next-token probabilities are shaped by complex learned patterns and context. But generation is not a guarantee of truth. Hallucinations can happen, so important claims should be verified. Modern AI assistants extend the base model with tools, retrieval, instructions, context, safety and other capabilities.

EPISODE 03 COMPLETE

Does ChatGPT Know or Does It Guess?